v4.6.0 Explainer

v4.6.0 Explainer — Slides 1-50

Deck: `docs/menopause_sns_gtex_enhanced_2026-05-07_v4.6.0.pptx` (50 slides, 13.33 × 7.50 in widescreen, dated 2026-05-07)

v4.6.0 changelog (layout + theme + slide-3 figure update; zero edits to audited prose): - **Layout fix:** 21 pictures across 18 slides repositioned via aspect-preserving contain-fit + horizontal letterbox (h ≤ 5.4”). Worst case in v4.5.5 was Saltiel ADRB3 slides 35/37/39/41 — figure forced to height 9.86” against a 7.50” canvas, cutting ~3.66” of figure below the slide bottom. - **Theme reskin:** the 13 v4.5.4-introduced “mirror” slides (4, 7, 28, 29, 31, 35, 37, 39, 41, 43, 47, 48, 49) used a flat minimal style (no title bar, tiny page-num) inconsistent with the ADRB2 originals. They are rebuilt to match the same Pattern A theme used throughout the deck: navy title bar (#1B2A4A) with white 22pt bold text; light-blue caption strip (#E8F4F8); light-gray footer bar (#F0F0F0). Title and caption text content preserved verbatim. - **Slide 3 figure (significance markers on the image):** `kenichi_inflammation_by_depot_menopause.png` regenerated by `R/next_steps_2026-05-07_slide3_significance_markers.R` with on-figure significance asterisks per (gene_set × depot) facet: HALLMARK Inflammatory subq * (p=0.044), Senescence subq *** (p=7.7e-4), Senescence visc ** (p=0.005); other facets `ns`. p-values shown as small text below each asterisk. Original analysis (`analysisA_tbl`) unchanged — only the plot rendering was extended with `geom_text` annotations. - **Footer version:** all 50 slides bumped from `v4.5.5` to `v4.6.0`. - **Audit chain:** all slide-text content (titles, captions, body bullets, table cells, audited prose, numerical claims, R-script methods references) carries forward byte-identical from v4.5.5. The Round-1+2+4 audit chain transfers verbatim. - **Patchers:** `scripts/make_pptx_2026-05-07_v4.6.0_layout_and_theme.py` (combined layout + theme + figure-replace + footer-bump) and `R/next_steps_2026-05-07_slide3_significance_markers.R` (figure regen). - **Verification:** 0 overflow shapes, 0 text-pic occlusions (vs 10 + 30 in v4.5.5); 50/50 footers say `v4.6.0`; 0/50 still say `v4.5.5`.

Predecessor (superseded for layout only): `docs/menopause_sns_gtex_enhanced_2026-05-04_v4.5.5.pptx` (kept on-disk for audit-trail traceability).

Predecessors: v4.5.1 (2026-04-22) → v4.5.2 (2026-05-01) → v4.5.3 (2026-05-01) → v4.5.4 (2026-05-03) → v4.5.5 (2026-05-04). v4.5.4 added 13 new analysis slides for ADRB3 mirrors / 50s-excluded sensitivity / signature-score stratification. v4.5.5 then incorporated a Round-4 audit (4 reviewers) that found two HIGH-severity factual errors (gene counts), one HIGH-severity statistical overstatement (the “ADRB3-fibrosis vanishes” language), and several MEDIUM cross-reference errors. v4.5.5 corrects all 9 issues. Audit chain status: **98 numerical claims verified in rounds 1-2 (zero mismatches); 26 R-script methods verified clean in round 4 (zero mismatches); 9 prose/cross-reference/hedging fixes applied between v4.5.4 and v4.5.5**. See the Revision history block at the bottom for the full ledger including Round 4. **Version lineage**: - In an earlier build, slides 1-46 contained v4.3.0 baseline content (2026-03-04) rescaled to the 13.33 × 7.50 in canvas; those slides have been removed in this revision, so the deck now opens directly with what were previously slides 47 onward. - Slides 1-14 (added in v4.4.0, 2026-04-12): Kenichi follow-up targeted analyses, glycolysis gene-node modeling, CIBERSORTx pipeline status. - Slides 15-22 (added in v4.4.1, 2026-04-12): CIBERSORTx HiRes differential expression results (both depots, 14 cell types). - Slides 23-37 (added in v4.5.0, 2026-04-21): thorough response to Kenichi’s 2026-04-21 email (OVX catecholamine resistance context, Q1–Q4 answers, updated cross-species concordance).

Purpose of this explainer: every slide is documented in detail — what it shows, what the data mean, where the numbers come from, and known caveats. Use this as the reference companion when reviewing the deck with Kenichi or Christoph, or when writing the manuscript.

Reading guide: the v4.5.4 deck is v4.5.3 plus 13 new slides. Sections 0 and 0.A below document everything new in v4.5.4 (read these first). Sections 1-3 below them are the inherited v4.5.3 prose **using v4.5.3 slide numbering (1-37)**; the slide-number map in §0.A tells you the corresponding v4.5.4 position for each v4.5.3 slide.

Section 0 — v4.5.4 additions (read this section first)

This section documents the 13 new slides added 2026-05-03 in response to collaborator-meeting feedback, plus the slide-number map from v4.5.3 → v4.5.4.

0.A Slide-number map: v4.5.3 → v4.5.4

v4.5.3 #	v4.5.5 #	Slide content
1	1	Section 1 — Update (2026-04-12)
2	2	Kenichi Follow-Up: Overview
3	3	Inflammation by Depot and Menopause Status
—	**4**	**NEW: HALLMARK vs cytokine inflammation methods clarification**
4	5	Aging vs Menopause: Sex-Interaction Analysis
5	6	ADRB2 Correlation Deep Dive
—	**7**	**NEW: ADRB3 correlation deep dive (mirror of v4.5.4 Slide 6)**
6	8	Human-Mouse Concordance v1 (historical)
7-14	9-16	Glycolysis sub-section + 2026-04-12 outputs summary
15	17	

v4.5.3 #	v4.5.5 #	Slide content
		CIBERSORTx HiRes section header
16-22	18-24	CIBERSORTx HiRes pipeline through BoxViolin grid
23-25	25-27	Section 3 header through Slide 25 (ADRB2 forest 50s in older)
—	**28**	**NEW: ADRB3 forest, 20-49 vs 50-79 (50s in older)**
—	**29**	**NEW: Sensitivity — 50s-included vs 50s-excluded older group**
26	30	Q1 Fisher-z table (ADRB2)
—	**31**	**NEW: Q1 mirror Fisher-z for ADRB3 (3 stratifications)**
27-29	32-34	Q2 / Q3 panels through Saltiel female subq (ADRB2)
—	**35**	**NEW: Saltiel ADRB3 female subcutaneous (n=233)**
30	36	Saltiel ADRB2 female visceral
—	**37**	**NEW: Saltiel ADRB3 female visceral (n=180)**
31	38	Saltiel ADRB2 male subcutaneous

v4.5.3 #	v4.5.5 #	Slide content
—	**39**	**NEW: Saltiel ADRB3 male subcutaneous (n=481)**
32	40	Saltiel ADRB2 male visceral
—	**41**	**NEW: Saltiel ADRB3 male visceral (n=407)**
33	42	Saltiel summary stats (ADRB2)
—	**43**	**NEW: Saltiel summary stats (ADRB3, 16 rows)**
34-36	44-46	Concordance heatmap + table + interpretation
—	**47**	**NEW: Section 4 — Menopause signature score stratification (distributions)**
—	**48**	**NEW: ADRB2 correlations stratified by senMayoLike score tertile**
—	**49**	**NEW: ADRB3 correlations stratified by senMayoLike score tertile**
37	50	Caveats and proposed next steps

0.B v4.5.4 Slide 4 — HALLMARK vs cytokine inflammation methods clarification

A text-only methods slide explaining the two inflammation measures used throughout the deck.

- **Kenichi cytokine score**: per-gene z-score then row-mean across a curated list of canonical pro-inflammatory cytokines (IL6, IL1B, TNF, CCL2, CXCL8, etc.). This is the headline measure because it is directly comparable to Kenichi's mouse panel.
- **HALLMARK_INFLAMMATORY_RESPONSE**: 200-gene MSigDB hallmark gene set (broader transcriptional inflammation signature: NF-κB targets, complement, leukocyte chemotaxis, etc.).

Where each is used: Slide 3 reports BOTH (subq cytokines p=0.14 trend UP; subq HALLMARK p=0.044 sig UP). The cross-species concordance table (now Slide 45) uses the cytokine score; if HALLMARK had been used instead, the iWAT inflammation row would shift from INDETERMINATE to CONCORDANT. This methods note makes that choice explicit on a slide of its own.

Manuscript recommendation: report both; lead with cytokine score for cross-species comparability with the mouse panel; present HALLMARK as a sensitivity check.

0.C v4.5.4 Slide 7 — ADRB3 correlation deep dive (mirror of Slide 6)

Two-image slide. Left: ADRB3 vs gene-set scores (4 panels: fibrosis / senescence / cytokine inflammation / lipolysis core), females, both depots. Right: ADRB3 vs CIBERSORT cell-type fractions. Asterisk significance markers throughout.

This is a direct mirror of Slide 6's ADRB2 panel for collaborator comparability: same panels, same axes, same significance markers; the only swap is ADRB2 → ADRB3 on the x-axis.

Source figures: `adrb3_correlation_panel.png`, `adrb3_vs_cibersort_fractions.png` (both in `figs/next_steps_2026-05-03_v4.5.4/`).

0.D v4.5.4 Slide 28 — Q1 mirror: ADRB3 forest, 20-49 vs 50-79 (50s in older)

Forest plot mirroring Slide 27 (ADRB2). Females only; broad stratification with 50s assigned to the older group. Each gene-set / gene / fraction target gets a 95% CI bar for both age strata, color-coded young vs older. Asterisk markers (* $p<0.05$) added to the right of each CI.

Headline finding: the ADRB3-vs-gene-set correlations in the broad stratification are more modest than ADRB2's. ADRB3 vs senescence: r_older=-0.21 (sig); ADRB3 vs fibrosis: r_older=-0.17 (NS at 0.05).

Source: `adrb3_age_stratified_forest_female.png`. Underlying TSV: `adrb2_adrb3_age_stratified_female_correlations.tsv`.

0.E v4.5.4 Slide 29 — Sensitivity: 50s-included vs 50s-excluded older group

The most important new analysis for the meeting. Females, both receptors, both depots. Compares the broad stratification (older = 50-79, n=155) against a sensitivity stratification (older = 60-79, n=79) that drops the 50-59 bin entirely. Older-group correlations are plotted side-by-side per (depot × receptor); orange = 50s-excluded, blue = 50s-included.

Headline findings:

Effect	50s-included (broad)	50s-EXCLUDED
ADRB2 × fibrosis (subq)	**r = −0.43**	**r = −0.48** (stronger)
ADRB3 × fibrosis (subq)	r = −0.17	**r = −0.06** (vanishes)
ADRB3 × fibrosis (peri 40-49 vs 50-59 only)	—	**r = −0.36** (sig)

Interpretation (statistically hedged after Round-4 audit): the ADRB2-fibrosis effect is robust and intensifies when 50s are removed. For ADRB3, the broad-analysis correlation drops from r=-0.17 to r=-0.06 when the 50s are removed, but a Fisher-z test for this attenuation yields **p=0.43 (not significant)**; the pure peri-window correlation r=-0.36 is numerically much larger than the 60-79-only correlation r=-0.06, but a Fisher-z test of those two yields **p=0.054 — marginally sub-threshold**. This pattern is suggestive that the ADRB3-fibrosis association may be concentrated in the perimenopausal transition, but it does not yet clear the conventional α=0.05 threshold; replication in a larger cohort is required.

Mechanistic reading (hypothesis only, not established): a possible interpretation is that ADRB3 is acutely impaired during the menopausal transition (50s) but recovers afterward, whereas ADRB2 impairment is lasting — different desensitization dynamics for the two receptors. This is a hypothesis suggested by the directional pattern; the data are not yet conclusive.

Source: `sensitivity_50s_excluded_forest_female.png`. TSV: `sensitivity_50s_excluded_fisherz.tsv` (broad-50s-excluded rows of the larger 3-stratification table).

0.F v4.5.5 Slide 31 — Q1 mirror: Fisher-z young-vs-older for ADRB3 (3 stratifications)

Text-based summary of the ADRB3 Fisher-z comparison across all three stratifications. Numerical contrast: 50s-included broad analysis (ADRB3-fibrosis r_older=-0.17, n=155); peri 50-59 only (r_older=-0.36, n=76); 50s-excluded (r_older=-0.06, n=79). The patterns are described in §0.E with the formal Fisher-z p-values (broad-incl vs broad-excl: p=0.43; peri vs broad-excl: p=0.054). This slide presents the case in tabular form on the deck for collaborator reference.

Full TSV: `adrb3_age_correlation_young_vs_older_fisherz.tsv` (12 rows × 11 columns; ADRB3 only across all 3 stratifications and key targets).

0.G v4.5.5 Slides 35, 37, 39, 41 — ADRB3 Saltiel-style scatter panels (4 mirrors)

Each slide mirrors its v4.5.3 ADRB2 counterpart (Slides 29-32 of v4.5.3 → Slides 34, 36, 38, 40 in v4.5.5) with ADRB3 substituted on the x-axis. The four panels per slide show ADRB3 vs Age / CCL2 / Fibrosis / Senescence. Asterisk markers in panel titles. Caption shows n, slope, R^2, p for each panel.

Headline R^2 values (v4.5.5 Slide 35, female subq, n=233) with Round-4 Williams' tests for dependent correlations (ADRB2 and ADRB3 measured in the same samples; r_ADRB2_ADRB3 = +0.287 was used):

Endpoint	ADRB2 R^2	ADRB3 R^2	Numerical winner	Williams t (p)
Age	0.021	0.047	ADRB3 (numerically)	t=+0.96 (p=0.34, NS)

Endpoint	ADRB2 R^2	ADRB3 R^2	Numerical winner	Williams t (p)
CCL2	0.090	0.027	ADRB2 (numerically)	t=−1.82 (p=0.069, NS)
Fibrosis	**0.174**	0.026	**ADRB2 (significantly)**	**t=−3.54 (p=0.0004)** ***
Senescence	0.076	0.093	ADRB3 (numerically)	t=+0.41 (p=0.68, NS)

Interpretation (statistically hedged after Round-4 audit): only the **fibrosis** dimension shows a statistically significant ADRB2-vs-ADRB3 difference (ADRB2 dominant, p=0.0004). The Age, CCL2, and Senescence numerical differences are directionally consistent with receptor specialization but are not statistically distinguishable at n=233 (all p > 0.06). The pattern of "ADRB2 owns inflammation/fibrosis; ADRB3 owns age/senescence" is suggestive but only one axis (fibrosis) is established at conventional significance. Lead manuscript framing on the fibrosis specialization; treat the other three axes as descriptive observations needing larger N for formal demonstration.

Source files: `saltiel_adrb3_scatters_{female,male}_{subcutaneous,visceral}.png`.

0.H v4.5.4 Slide 43 — Saltiel summary stats for ADRB3 (16 rows)

Mirror of Slide 42 (ADRB2). Same 16 rows (4 sex×depot subsets × 4 endpoints) but for ADRB3. The slide presents the head-to-head comparison at a glance: which receptor wins on which endpoint, in which subset.

Bottom-line text on the slide (v4.5.5 Round-4-hedged): *"ADRB2's dominance on the fibrosis axis is statistically significant (Williams t=-3.54 for dependent correlations, p=0.0004). The other three apparent specializations (Age, CCL2, Senescence) are directionally consistent but not statistically distinguishable at n=233. Larger cohort needed to formally establish receptor specialization on those axes."*

Source TSV: `saltiel_style_regression_stats_adrb3.tsv` (16 rows; same schema as v4.5.3's `saltiel_style_regression_stats.tsv`).

0.I v4.5.4 Section 4 — Menopause signature score stratification (Slides 47-49)

A new section motivated by collaborator suggestion to stratify samples by **transcriptional menopause state** rather than chronological age bin.

Two signatures are computed per sample (females only, separately per depot):

- **senMayoLike v2** (`references/menopause_signature_senMayoLike_v2.tsv`, **13 genes** total, 12 UP + 1 DOWN): score = mean[z(UP genes)] − mean[z(DOWN genes)]. Signed — high = "more post-menopause-like". Note: this is a curated subset designed for SenMayo-like menopause-signature use, distinct from the broader published SenMayo gene set; gene counts are smaller because of stringent filtering for adipose-relevant biology and avoidance of lncRNA / pseudogene / keratin / blood-heavy markers.
- **curated v1** (`references/menopause_signature_curated.tsv`, **72 genes used** out of 81 rows in the file; the script uses only rows with `keep=TRUE`): score = mean[z(all kept genes)]. Unsigned magnitude.

Slide 47 — Score distributions and tertile cutoffs

Scatter of senMayoLike v2 (x) vs curated (y) per sample, faceted by depot, colored by age-bin menopause group. Dashed lines = tertile cutoffs per signature × depot.

Concordance with age-bin grouping (computed in Round-4 audit, by depot × signature):

Depot	Signature	% "pre" by age in T3_high	% "post" by age in T1_low
Subcutaneous	senMayoLike v2	15.4% (12/78)	25.3% (20/79)
Subcutaneous	curated	19.2% (15/78)	(similar)
Visceral	senMayoLike v2	29.2% (21/72)	27.8% (15/54)
Visceral	curated	29.2% (21/72)	(similar)

Cohen's kappa between age-bin and score tertile (both as 3-level categorical): **0.169 in subcutaneous, 0.100 in visceral** — both fall in the "slight agreement" range (Landis & Koch 1977 cutoffs: <0.20 = slight; 0.21-0.40 = fair). The two groupings are largely discordant.

Implication (Round-4-hedged): this confirms that transcriptional score and chronological age capture genuinely different variance, and **justifies using score-tertile grouping as a sensitivity analysis** in the manuscript. It does NOT yet support replacing age-bin as the primary stratification — that would require outcome-validation showing stronger effect sizes within score-based groupings, which has not been performed.

Source: `menopause_score_distribution.png`. Concordance table: `score_tertile_vs_age_bin_concordance.tsv`.

Slide 48 — ADRB2 correlations stratified by senMayoLike score tertile

Forest plot of ADRB2 ↔ standard target panel (inflammation cytokines, HALLMARK, fibrosis, senescence, TGFβ/ALK7, lipolysis core, CA-resistance) within each of three tertiles (T1 = lowest score / "pre-like"; T3 = highest / "post-like"). Females only.

Reading: if ADRB2-fibrosis (and similar) correlations are progressively more negative across T1 → T2 → T3, this would corroborate the "menopause-driven β-AR desensitization" hypothesis at the transcriptional-state level — independent of age-bin.

Source: `score_tertile_correlations_adrb2_senmayo.png`. Curated-signature counterpart in `score_tertile_correlations_adrb2_curated.png`. Stats: `score_tertile_correlations_both_receptors.tsv`.

Slide 49 — ADRB3 correlations stratified by senMayoLike score tertile

Same as Slide 48 but for ADRB3. Mirror plot enables direct receptor comparison within score tertiles. **Of particular interest**: does the ADRB3-fibrosis effect that was 50s-specific in the age-bin analysis (Slide 29) reproduce as a T2/T3-specific effect in the score-based analysis? This would either corroborate the "ADRB3 effect is concentrated in the menopausal transition" hypothesis or argue for a methodological artifact.

Source: `score_tertile_correlations_adrb3_senmayo.png`. Curated counterpart in `score_tertile_correlations_adrb3_curated.png`.

0.J v4.5.4 significance markers — applied throughout the new slides

All v4.5.4-introduced figures use the standard convention:

p-value	Marker
$p < 0.001$	`***`
$p < 0.01$	`**`
$p < 0.05$	`*`
$p \geq 0.05$	`ns` (or omitted)
missing / undefined	`n/a`

For boxplots and forest plots: asterisks shown next to the bar / CI. For scatter regressions: asterisk placed in the panel title after the gene/endpoint label. For the tabular Q1 Fisher-z slide (Slide 31): asterisks inline next to the delta values.

Note: existing v4.5.3 slides retain their original look (no asterisk overlay). Only the 13 v4.5.4-new slides carry the unified marker convention. Adding markers to legacy slides is an open improvement noted in the audit log; not yet done because it would require regenerating PNGs that are not regression-critical for the meeting.

Section 1-3 reference — v4.5.3 slide-by-slide content (predecessor numbering)

The prose below refers to the **v4.5.3 deck** with slides numbered 1-37. To find the corresponding v4.5.4 slide number for any reference below, use the slide-number map in §0.A above.

Section 1 — Kenichi Follow-Up & Glycolysis Gene-Node Modeling (Slides 1-14, v4.4.0)

Slide 1 — Section Header: Update (2026-04-12)

Transition slide marking the start of the v4.4.0 content. The subtitle establishes the three work streams covered in slides 2-14: the targeted follow-up to Kenichi’s 2026-03-27 email, glycolysis pathway gene-node modeling, and the then-in-progress CIBERSORTx pipeline. No analytical content on this slide — its role is purely structural. In the previous (v4.5.1 original) build this slide separated the v4.3.0 baseline content from the 2026-04-12 additions; in the current revision the baseline content has been removed, so this slide now opens the deck.

Slide 2 — Kenichi Follow-Up: Overview

A four-box summary of the targeted follow-up addressing Kenichi's 2026-03-27 email. Each box labels one of his four points and the specific analysis we designed in response:

- **Point 1 (inflammation depot specificity)**: Kenichi noted inflammation appears UP in subcutaneous but NOT in visceral — opposite to his OVX mouse model. We tested this with a formal depot × menopause interaction test (females only, pre vs post, adjusted for SMCENTER).
- **Point 2 (baseline transcriptome limitation)**: We agreed with Kenichi that bulk baseline transcriptomic data cannot prove SNS activity. The response pointed to the then-in-progress CIBERSORTx HiRes analysis for cell-type resolution.
- **Point 3 (aging vs menopause confound)**: Kenichi observed that male and female trends appear parallel in some gene sets (this was Slide 43 of the older 83-slide v4.3.0 deck; that baseline content has since been removed, so the slide is no longer accessible from this deck). We fit a sex × age_group interaction model to isolate female-specific effects beyond generic aging.
- **Point 4 (ADRB2 correlations)**: Kenichi noted ADRB2 is negatively correlated with inflammation/fibrosis, consistent with catecholamine resistance. We performed a deep correlation analysis with CIBERSORT fractions and produced a human-mouse concordance table.

The bottom panel summarizes five key findings from the follow-up, including the depot-specific inflammation pattern, senescence UP in both depots, the aging-vs-menopause interaction results, and an early version of the concordance table (8/14 concordant). Note: this concordance "8/14" figure is from the v4.4.0-era table and was later superseded in v4.5.0 after the concordance regex bug was fixed on 2026-04-22 (see Slide 35).

Source files: `R/next_steps_2026-04-12_kenichi_followup_targeted.R`.

Slide 3 — Inflammation by Depot and Menopause Status (Point 1)

Left panel: a faceted boxplot showing cytokine inflammation, HALLMARK inflammatory response, and senescence gene-set scores by menopause proxy group (pre, peri, post) within each depot (subcutaneous vs visceral). Right panel: bulleted interpretation.

Key numbers (from `kenichi_inflammation_depot_comparison.tsv`): - Subcutaneous inflammation cytokines: estimate = +0.179, p = 0.138 (UP in post, trend) - Subcutaneous HALLMARK inflammatory: estimate = +0.124, p = 0.044 (significant UP in post) - Subcutaneous senescence: estimate = +0.355, p = 0.00077 (strong UP in post) - Visceral cytokine inflammation: estimate = −0.182, p = 0.131 (DOWN trend in post, opposite direction to subq) - Visceral HALLMARK inflammatory: estimate = +0.007, p = 0.920 (essentially flat) - Visceral senescence: estimate = +0.330, p = 0.005 (significant UP) - Depot × menopause interaction for cytokines: estimate = −0.511, p = 0.003 (formally significant depot-specific divergence)

Interpretation: the cross-species discordance is real at the level of inflammation in visceral adipose — humans show a non-significant DOWN trend while OVX mice show UP. Possible explanations include: (1) human GTEx "visceral omentum" is not anatomically analogous to mouse gonadal fat; (2) species-specific immune infiltration patterns; (3) steady-state post-menopausal tissue differs from acute OVX.

Slide 3 subtitle (added in v4.5.3): the formal depot × menopause interaction p-value (cytokines: p = 0.003) is now visible in the slide subtitle so a viewer of the slide alone can see the headline statistical evidence for depot-specific divergence without needing to consult the explainer.

Source files: `GTEx_v10_AT_analysis_out/tables/next_steps_2026-04-12_kenichi_followup/kenichi_inflammation_depot_comparison.tsv`, `figs/next_steps_2026-04-12_kenichi_followup/kenichi_inflammation_by_depot_menopause.png`.

Slide 4 — Aging vs Menopause: Sex-Interaction Analysis (Point 3)

Left panel: multi-faceted trajectory plot showing gene-set score means by GTEx age bin, colored by sex, separate panels for subcutaneous vs visceral. Right panel: text summary of the sex × age interaction model results.

The statistical model was `score ~ SMCENTER + sex + age_group + sex:age_group` over two age contrasts: 40-49 vs 60-69 (broad menopause-adjacent window) and 40-49 vs 50-59 (narrower peri-menopausal transition).

Key findings for 40-49 vs 60-69 subcutaneous (from `kenichi_aging_vs_menopause_interaction.tsv`): - Inflammation cytokines: male p = 2.4e-4 (highly sig), female p = 0.13 (NS) → males show stronger aging effect - HALLMARK inflammatory: male p = 0.019, female p = 0.11 → males stronger - Senescence: male p = 2.4e-4, female p = 0.046 → both sexes show effect - Lipolysis core: male p = 4.0e-4 (DOWN), female p = 0.17 → aging effect in both sexes - TGFβ/ALK7 axis: male p = 0.09, female p = 0.66 (NS)

Peri-menopausal window (40-49 vs 50-59) visceral shows different picture with significant female-specific interactions: - TGFβ/ALK7 axis: interaction p = 0.002 (females UP +0.357, males −0.101) - Catecholamine resistance: interaction p = 0.003 (females +0.187, males −0.104) - Fibrosis: interaction p = 0.012 (females +0.238, males −0.051) - HALLMARK inflammatory: interaction p = 0.020 (females +0.151, males −0.065)

Bottom line: broad aging trends (40-49 vs 60-69) are largely parallel between sexes in subcutaneous (no formal female-specific excess). Female-specific effects emerge most clearly in visceral adipose during the narrower peri-menopausal transition. Manuscript framing should lead with the visceral peri-menopausal findings rather than overclaiming the broader age window.

Slide 4 conclusion (revised in v4.5.3): the on-slide conclusion now reads "Conclusion: in this 40-49 vs 60-69 window, aging trends are largely shared between sexes. Formal female-specific effects (TGF-b/ALK7, CA-resistance) appear in the visceral peri-menopausal window." The previous v4.5.2 wording attributed TGF-b/ALK7 and CA-resistance as female-specific without flagging that those were significant only in the visceral 40-49 vs 50-59 window not shown on this slide; the round-2 audit caught this and the conclusion has been corrected.

Source files: `kenichi_aging_vs_menopause_interaction.tsv`, `kenichi_aging_vs_menopause_interaction_40to60_window.tsv`, `figs/.../kenichi_aging_vs_menopause_trajectories.png`.

Slide 5 — ADRB2 Correlation Deep Dive (Point 4)

Two-image slide: left panel is the ADRB2 ↔ gene-set-score correlation panel (scatter plots for top 4 gene sets, both depots); right panel is ADRB2 vs CIBERSORT cell fractions. These figures anchor the conclusion that ADRB2 acts as an inverse marker of inflammatory/fibrotic burden in subcutaneous tissue and is positively associated with adipocyte fraction.

Headline correlations (from `kenichi_adrb2_correlations_detail.tsv`): - Subcutaneous, female: ADRB2 × fibrosis r = −0.42, p = 3.3e-11 (strongest negative) - Subcutaneous, male: ADRB2 × fibrosis r = −0.46, p = 4.5e-26 (strongest male) - Subcutaneous, both sexes: ADRB2 × cytokine inflammation r ≈ −0.24 (highly sig) - Visceral, male: ADRB2 × cytokine inflammation r = +0.11, p = 0.023 (POSITIVE — opposite direction to subq) - Visceral, female: ADRB2 × cytokine inflammation r = +0.07, p = 0.35 (NS, but directionally positive) - Subcutaneous, female: ADRB2 × adipocyte fraction r = +0.47, p = 3.8e-14 (strong positive) - Subcutaneous, male: ADRB2 × adipocyte fraction r = +0.32, p = 5.3e-13

The depot-specific reversal of the ADRB2 × inflammation relationship in visceral is the most noteworthy new finding here. It suggests that the catecholamine-resistance interpretation of ADRB2 is depot-dependent and should not be generalized without caveat.

Source files: `kenichi_adrb2_correlations_detail.tsv`, `figs/.../kenichi_adrb2_correlation_panel.png`, `figs/.../kenichi_adrb2_vs_cibersort_fractions.png`.

Slide 6 — Human-Mouse Concordance: GTEx vs OVX Model (early version)

A 14-row concordance table comparing expected OVX mouse direction against human GTEx direction for each phenotype × depot combination. Each row includes the significance of the human result and a concordance label.

Status note (IMPORTANT): This v4.4.0-era concordance table was built before the OVX 042126 email arrived. It uses “8/14 concordant” as the summary, treats visceral inflammation/fibrosis as DISCORDANT (same as v4.5.0), and lists ADRB2 direction as DISCORDANT based on the pre-vs-post comparison (which showed a slight UP trend in humans, opposite to the mouse DOWN expectation).

The **current** authoritative concordance table is on Slide 35 (v4.5.0) and reflects a bug fix applied 2026-04-22 that correctly marks iWAT inflammation/fibrosis/senescence as INDETERMINATE rather than CONCORDANT, because Kenichi’s OVX 042126 pptx only shows pWAT measurements for those phenotypes. When presenting this deck, it is best to treat Slide 6 as historical context and direct attention to Slide 35 for the final interpretation.

Source file: `kenichi_human_mouse_concordance.tsv` (2026-04-12 version — superseded by `concordance_with_ovx_042126.tsv` for v4.5.0).

Slide 7 — Section Header: Glycolysis Gene-Node Analysis

Transition slide marking the start of the glycolysis sub-section (slides 8-12). The work here modeled 221 individual glycolysis pathway genes (HALLMARK_GLYCOLYSIS members plus 32 named candidate genes) both at baseline and after CIBERSORT-based composition adjustment, in both depots.

Slide 8 — Glycolysis Gene-Node Trajectories: Subcutaneous

A wide (12.7 in) faceted trajectory plot showing VST expression of the top glycolysis genes by GTEx age bin, stratified by sex, for subcutaneous adipose. Each facet is a single gene; the x-axis is age bin; the y-axis is VST; male and female trajectories are overlaid.

Purpose: to visually document which glycolysis genes show age-dependent trajectories and whether those trajectories differ by sex. Genes such as PFKFB3, HK1, GAPDH, and PGK1 are typical highlights. The trajectories show a directional bias toward decline in post-menopause females; in subcutaneous tissue, 38 of 221 modeled genes (17.2%) reach nominal $p<0.05$ with a negative post-vs-pre coefficient (formal effect-size estimates and forest plot are on Slide 10). Visual trajectories should be interpreted alongside Slide 10 rather than as standalone evidence of reduced aerobic-glycolytic capacity.

Source files: `R/next_steps_2026-04-12_glycolysis_gene_nodes.R`, `figs/next_steps_2026-04-12_glycolysis_gene_nodes/glycolysis_gene_node_trajectories_subcutaneous.png`.

Slide 9 — Glycolysis Gene-Node Trajectories: Visceral

Same content structure as Slide 8 but for visceral omentum. Visceral trajectories are generally flatter than subcutaneous: 7 of 221 genes (3.2%) reach nominal $p<0.05$ with a negative post-vs-pre coefficient, versus 38/221 (17.2%) in subcutaneous. The predominant direction in visceral remains negative (151/221 genes, 68.3%) — even slightly more so than in subcutaneous (120/221, 54.3%) — so the contrast with subcutaneous is one of statistical power, not direction reversal. (Earlier wording in v4.5.1 said “opposite direction”; that was incorrect and was revised after the 2026-04-30 audit. See Revision history.)

Source file: `figs/next_steps_2026-04-12_glycolysis_gene_nodes/glycolysis_gene_node_trajectories_visceral.png`.

Slide 10 — Glycolysis Gene-Node Effects: Post vs Pre (Forest Plots)

Side-by-side forest plots: left = subcutaneous, right = visceral. Each forest plot shows the post-vs-pre menopause coefficient (and 95% CI) for each glycolysis gene, estimated by the base LM (~

`SMCENTER + menopause_group`). The forest format makes it easy to see which genes move UP vs DOWN in post-menopause and which have CIs that exclude zero.

Interpretation: In subcutaneous, a meaningful minority of glycolysis genes show negative post-vs-pre coefficients (DOWN in post) at nominal significance. In visceral, coefficients cluster more tightly around zero.

Source files: `tables/next_steps_2026-04-12_glycolysis_gene_nodes/glycolysis_gene_node_lm_stats_subcutaneous.tsv`, `..._visceral.tsv`. Forest PNGs: `figs/next_steps_2026-04-12_glycolysis_gene_nodes/glycolysis_gene_node_forest_subcutaneous.png`, `..._visceral.png`.

Slide 11 — Glycolysis: Effect Attenuation After Composition Adjustment

Two scatter plots (left subq, right visc): X-axis = base-model post-vs-pre coefficient; Y-axis = CIBERSORT-adjusted coefficient. Points on the diagonal indicate effects robust to composition adjustment (cell-intrinsic); points pulled toward the X-axis ($Y \approx 0$) indicate effects driven by cell-composition shifts.

Interpretation: Most glycolysis genes show modest attenuation along the Y-axis after CIBERSORT adjustment, meaning both cell-intrinsic and composition-driven contributions exist. A small number of genes collapse strongly toward the X-axis (pure composition effects) and a small number retain almost identical coefficients (pure intrinsic effects).

Source files: `figs/next_steps_2026-04-12_glycolysis_gene_nodes/glycolysis_gene_node_effect_attenuation_subcutaneous.png`, `..._visceral.png`.

Slide 12 — Glycolysis Genes vs CIBERSORT Fractions: Correlation Heatmaps

Two correlation heatmaps (left subq, right visc) showing the Pearson correlation between each glycolysis gene's VST expression and each CIBERSORT cell-type fraction. This tells us which cell types "carry" the expression of each gene — for example, genes with strong positive correlation to adipocyte fraction are likely adipocyte-expressed; genes that correlate with macrophage fraction are likely immune-cell-expressed.

Interpretation: The heatmap reveals that subset of glycolysis genes (e.g., ALDOA, GAPDH) have fairly uniform expression across cell types, while others show clear cell-type bias (e.g., preferential ASPC or macrophage expression).

Source files: `figs/next_steps_2026-04-12_glycolysis_gene_nodes/glycolysis_fraction_correlation_heatmap_subcutaneous.png`, `..._visceral.png`.

Slide 13 — CIBERSORTx HiRes Pipeline: Status and Next Steps (as of v4.4.0)

A three-box layout describing the state of the CIBERSORTx pipeline at v4.4.0 generation time (approximately mid-afternoon 2026-04-12). At that point, Docker images were installed, reference was identified (GSE176171), inputs were prepared (refsample 2.3 GB, mixtures 412 MB subq / 338 MB visc), but Docker execution was still pending due to a token authentication issue.

Status note: By v4.4.1 (same day, later), the Docker execution succeeded after token regeneration, HiRes ran for both depots (subq 47 min, visc 41 min), and the DE analysis completed. So this slide is **historical** and the reader should proceed to Slide 15 onward for actual HiRes results.

Expected outputs described on this slide: 1. CIBERSORTx Fractions with S-mode batch correction + 100 permutations 2. CIBERSORTx HiRes per-sample per-cell-type expression matrices 3. Downstream DE on the imputed expression (pre/peri/post menopause + male comparison)

Source files: `R/next_steps_2026-04-12_prepare_cibersortx_inputs.R`, `scripts/run_cibersortx_pipeline_2026-04-12.sh`, `R/next_steps_2026-04-12_cibersortx_hires_de_analysis.R`.

Slide 14 — 2026-04-12 Update: New Outputs Summary

Four-box directory of all outputs generated in the 2026-04-12 Update session: - Box 1: Kenichi follow-up — 1 R script, **6 tables visible on the slide** (inflammation depot comparison, 2 aging-vs-menopause tables, 2 ADRB2 correlation tables, concordance) — note: a 7th table (`all_scores_both_depots.tsv`) exists on disk but was not listed in the slide's box; in v4.5.3 the slide header was corrected from "Tables (7):" to "Tables (6):" to match what is shown. - Box 2: Glycolysis gene-nodes — 1 R script, 6 tables, 8 figures. - Box 3: CIBERSORTx — 3 scripts (prepare, run, DE) plus prepared inputs (3.1 GB). At v4.4.0 time this was still pending. - Box 4: Email draft — `docs/reply_to_kenichi_2026-04-12.md` (the April 12 response email that was never sent; its content was later consolidated into `reply_to_kenichi_2026-04-21_quick.md` after the April 21 session). The version label on this box was clarified in v4.5.3 from "Deck version: v4.4.0 (partial; CIBERSORTx pending)" to "Snapshot label at email draft time (2026-04-12): v4.4.0 partial" so a reader does not mistake it for the current deck version.

Serves as a quick directory for anyone navigating the 2026-04-12 artifacts.

Section 2 — CIBERSORTx HiRes Cell-Type-Resolved DE (Slides 15-22, v4.4.1)

Slide 15 — Section Header: CIBERSORTx HiRes

Section opener for the CIBERSORTx HiRes DE results. The subtitle emphasizes three key design choices: both depots analyzed, 14 cell types resolved, and the female menopause proxy is paired with a male comparison cohort to control for non-menopause aging effects.

Slide 16 — CIBERSORTx Pipeline: What Was Done

A six-box layout describing the end-to-end CIBERSORTx workflow:

- **Reference**: GSE176171 single-cell adipose atlas (human WAT), 14 cell types annotated.
- **Fractions**: Signature matrix built from the sc reference inside CIBERSORTx Docker, with S-mode batch correction and 100 permutations for confidence estimation.
- **HiRes Imputation**: Sample-level cell-type-specific expression inferred from bulk RNA, window=20, 10 threads per job. This is the key step — it converts bulk RNA into approximate per-cell-type expression for each GTEx sample.
- **Differential Expression**: Per-cell-type limma models, testing post vs pre menopause in females, with a male cohort for comparison.
- **Subcutaneous cohort**: 714 samples, 18,359 genes, 14 cell types.
- **Visceral cohort**: 587 samples, 18,359 genes, 14 cell types.

Key technical clarifications for the reader: - `window=20` was raised from the default 10 to exceed the threshold `window > (cell_types + 1)` required for 14 cell types. - For the HiRes step, we used the pre-adjusted mixture/signature from the Fractions step and disabled S-mode (which is only needed at Fractions step); this resolved an initial HiRes error.

Slide 17 — CIBERSORTx HiRes DE: Subcutaneous (Post vs Pre, Females)

A 14-row table showing per-cell-type differential expression counts for post-vs-pre menopause in subcutaneous females, sorted by total significant gene count. Columns: cell type, genes tested, UP count, DOWN count, total significant.

Top findings: - **Adipocyte**: 9,697 genes tested; **221 significant** (79 UP, **142 DOWN**). The asymmetric DOWN pattern is the single most striking finding — it suggests cell-intrinsic downregulation of the adipocyte transcriptome in post-menopausal subcutaneous tissue. - **ASPC (Adipose Stem/Progenitor Cells)**: 152 significant (112 UP, 40 DOWN) — strong UP bias, consistent with progenitor-cell activation post-menopause. - **Macrophage**: 128 significant (53 UP, 75 DOWN) — balanced. - **Smooth muscle cell (SMC)**: 138 (64 UP, 74 DOWN). - **Endothelial**: 124 (83 UP, 41 DOWN). - **Pericyte**: 0 significant DE genes — likely reflects poor imputation quality for pericyte in subcutaneous depot (noted as an open technical risk).

Interpretation note in the slide caption: "Adipocyte shows asymmetric DOWN (142 vs 79 UP), suggesting cell-intrinsic downregulation after menopause. ASPC and T cell show strong UP bias."

Filter applied: |logFC| > 0.5 AND FDR < 0.05 (applied in the summary; the raw DE tables contain all genes passing FDR < 0.05 regardless of logFC, including near-zero-logFC imputation artifacts).

Source files: `R/next_steps_2026-04-12_cibersortx_hires_de_analysis.R`, DE tables under `tables/next_steps_2026-04-12_cibersortx_hires/` (28 tables per depot).

Slide 18 — CIBERSORTx HiRes DE: Visceral (Post vs Pre, Females)

Same table structure as Slide 17 but for visceral omentum. Key contrasts with subcutaneous:

- **Endothelial** becomes the #1 DE cell type in visceral (141 significant: 84 UP, 57 DOWN).
- **Macrophage** is consistent across depots (130 in visc vs 128 in subq).
- **Adipocyte** is demoted to #3 in visceral (115 significant: 76 UP, 39 DOWN) — and the direction flips! In subq adipocyte is DOWN-biased (142 vs 79), while in visc it is UP-biased (76 UP, 39 DOWN). This is another depot-specific sign flip.

- **Mast cell** in visceral shows a striking DOWN-only pattern (3 UP, 26 DOWN) — unusual enough to flag as either biologically interesting or an imputation artifact warranting further investigation.

Interpretation note in the slide caption: “Endothelial becomes the top DE cell type in visceral. Macrophage consistent across depots. Mast cell shows striking DOWN-only pattern (3 up, 26 down).”

Slide 19 — Adipocyte: Post vs Pre Menopause (HiRes-Resolved)

Two volcano plots side by side: left = subcutaneous adipocyte, right = visceral adipocyte. Each volcano plots −log10(FDR) vs logFC for every gene in the adipocyte-imputed expression matrix. Genes above the horizontal FDR=0.05 threshold and outside |logFC|>0.5 are colored as significant.

Visual interpretation: - Subcutaneous adipocyte volcano shows the asymmetric DOWN pattern: a thicker tail of significant genes to the left (negative logFC) than to the right (positive logFC). - Visceral adipocyte volcano is more symmetric, with slight bias toward UP.

The comparison between the two volcanos visually reinforces the depot-specific direction difference noted on Slides 17-18.

Source files: `figs/next_steps_2026-04-12_cibersortx_hires/Volcano_subcutaneous_adipocyte_post_vs_pre_female.png`, `Volcano_visceral_adipocyte_post_vs_pre_female.png`.

Slide 20 — Macrophage: Post vs Pre Menopause (HiRes-Resolved)

Same layout as Slide 19 but for macrophage. Macrophage DE counts are similar between depots (128 subq, 130 visc), and the volcano shapes reflect this consistency. Together with Slide 19 this establishes that adipocyte and macrophage are both substantial contributors to the bulk signal.

Source files: `figs/next_steps_2026-04-12_cibersortx_hires/Volcano_subcutaneous_macrophage_post_vs_pre_female.png`, `Volcano_visceral_macrophage_post_vs_pre_female.png`.

Slide 21 — ASPC & Endothelial: Post vs Pre (Subcutaneous, HiRes)

Two volcano plots for subcutaneous ASPC (left) and endothelial (right). ASPC's UP-biased volcano (112 UP, 40 DOWN) is clearly visible. Endothelial in subcutaneous shows a moderate UP bias (83 UP, 41 DOWN). These are the second- and third-ranked cell types by total DE count in subcutaneous after adipocyte.

Source files: `figs/next_steps_2026-04-12_cibersortx_hires/Volcano_subcutaneous_aspc_post_vs_pre_female.png`, `Volcano_subcutaneous_endothelial_post_vs_pre_female.png`.

Slide 22 — Cell-Type-Specific Expression of Key Genes (Subcutaneous)

A 2×3 grid of box/violin plots showing per-sample cell-type-resolved expression across menopause proxy groups (pre/peri/post) in subcutaneous females. The six panels (verified by MD5 against the embedded images during the 2026-05-01 round-2 audit) are:

1. **Adipocyte / ADIPOQ** — adipokine, classic adipocyte marker.
2. **Adipocyte / LEP** — leptin, adipocyte energy-balance marker.
3. **Adipocyte / ESR1** — estrogen receptor α in adipocytes (the menopause-relevant receptor).
4. **SMC / ESR1** — estrogen receptor α in smooth muscle cells (vasculature).
5. **Macrophage / CD68** — pan-macrophage marker.
6. **Monocyte / IL6** — pro-inflammatory cytokine in monocytes.

Purpose: to visualize concrete per-gene patterns in the HiRes output — which genes show clean menopause-associated changes, which are noisy, and whether the direction of change matches expectations from the bulk analysis. Note: the panel layout intentionally focuses on adipokines + ESR1 + macrophage/monocyte inflammation rather than ADRB2 directly, since ADRB2 cell-type-specific resolution was deferred to a follow-up analysis (see Slide 37 next steps).

Source files (from `figs/next_steps_2026-04-12_cibersortx_hires/`): - `BoxViolin_subcutaneous_adipocyte_ADIPOQ_menopause.png` - `BoxViolin_subcutaneous_adipocyte_LEP_menopause.png` - `BoxViolin_subcutaneous_adipocyte_ESR1_menopause.png` - `BoxViolin_subcutaneous_smc_ESR1_menopause.png` - `BoxViolin_subcutaneous_macrophage_CD68_menopause.png` - `BoxViolin_subcutaneous_monocyte_IL6_menopause.png`

Section 3 — Response to Kenichi 2026-04-21 (Slides 23-37, v4.5.0)

This section was added in v4.5.0 in response to Kenichi's 2026-04-21 email (`references/042126_Email-from-Kenichi.txt`) and the attached mouse-data PPTX (`references/Catecholamine_resistance_in_OVX_042126.pptx`, 9 slides). Kenichi shared new OVX mouse data showing Adrb3 DOWN in both pWAT and iWAT (Adrb1 also DOWN in iWAT, more pronounced in iWAT), and inflammation/fibrosis/senescence UP in OVX pWAT. He posed four explicit questions, answered across Slides 24-36 below.

Slide 23 — Section Header: Response to Kenichi (2026-04-21)

Section opener. The subtitle previews the four work items to be addressed: age-stratified ADRB2 analyses, sex-specific decline tests, Saltiel Fig 8-style scatter panels, and cross-species concordance update.

Slide 24 — Recap: your OVX 042126 findings

Two-box layout. Left box ("What you showed and why it matters for our human GTEx analysis") summarizes Kenichi's key OVX observations: - Slide 1 of his PPTX: Adrb3 DOWN in both pWAT (visceral) and iWAT (subcutaneous); Adrb1 also DOWN in iWAT (the Adrb1 panel is visible in the embedded figures on his Slide 1, but the gene name itself does not appear in the slide's text labels — verified by python-pptx text extraction during the 2026-04-30 audit). - The reduction is MORE PRONOUNCED in iWAT. - Slide 2 of his PPTX: inflammation, fibrosis, senescence markers UP in OVX pWAT (similar trend in iWAT but NOT shown in his slides). - Hypothesis: estrogen deficiency → chronic SNS activity → catecholamine resistance (reduced β-AR signaling). - Context: Prog Lipid Res 2009 states ADRB1 AND ADRB2 (not ADRB3) are the primary lipolytic β-AR subtypes in human adipose.

Right box ("Your four questions") enumerates Kenichi's explicit asks: - Q1: Is ADRB2 – inflammation/fibrosis/senescence negative correlation STRONGER in older women vs younger women? - Q2: Is ADRB2 reduced in subcutaneous older women, and is the reduction more pronounced in women than men? - Q3: Reproduce Saltiel Fig 8-style scatter for ADRB2 vs age, CCL2, fibrosis, senescence in women. - Q4: Suggest best visualization approach.

This slide sets up the answers in Slides 25-33 (Q1-Q3) and Slide 36 (Q4).

Slide 25 — Q1: ADRB2 correlations in females — 20-49 vs 50-79 (forest plot)

A forest plot showing Pearson r (with 95% CI from Fisher-z) between ADRB2 and each gene-set target (inflammation cytokines, HALLMARK inflammation, fibrosis, senescence, TGFβ/ALK7, CA-resistance, lipolysis core, etc.), stratified by age: 20-49 (younger females) vs 50-79 (older females), for both depots.

The forest plot answer to Q1: the point estimates for older females are NOT systematically more negative than for younger females. For some targets (e.g., subcutaneous fibrosis) older is slightly more negative; for others (e.g., subcutaneous CCL2 and HALLMARK inflammation) older is slightly LESS negative. The 95% CIs overlap broadly.

Key statistic on the slide caption: "Fisher-z tests for young vs older female correlation magnitudes are all non-significant at $\alpha=0.05$. Min p = 0.062 for visceral TGFβ/ALK7 axis, where older females show a STRONGER POSITIVE correlation — opposite direction to catecholamine-resistance hypothesis."

This is an intellectually honest negative result: Kenichi's hypothesis (older women show stronger negative ADRB2-inflammation correlation) is NOT supported by the GTEx data.

Source files: `R/next_steps_2026-04-21_adrb2_age_stratified_saltiel_style.R`, `figs/next_steps_2026-04-21_adrb2_age_stratified/adrb2_age_stratified_forest_female.png`, `tables/.../adrb2_age_correlation_young_vs_older_fisherz.tsv`.

Slide 26 — Q1: Fisher-z comparison (older vs younger females) — top gene sets

An 18-row table showing for each (depot, target) pair: n_young, r_young, n_older, r_older, r_delta (older minus younger), Fisher-z p, and whether "older more negative" is TRUE.

Summary row-by-row: - Subcutaneous fibrosis: r_young=-0.379, r_older=-0.433, r_delta=-0.054, p=0.647 → YES older more negative, but NS. - Subcutaneous senescence: r_young=-0.315, r_older=-0.255, r_delta=+0.060, p=0.644 → NO (older LESS negative). - Subcutaneous CCL2: r_young=-0.410, r_older=-0.245, r_delta=+0.165, p=0.189 → NO (older LESS negative; largest directional counter-example). - Subcutaneous TGFB1: r_young=-0.237, r_older=-0.381, r_delta=-0.144, p=0.258 → YES older more negative, NS. - Visceral TGFβ/ALK7 axis (context from fisherz TSV, off-table): r_young=+0.036, r_older=+0.314, p=0.062 → older shows STRONGER POSITIVE correlation.

Slide caption summary: "Net result: no formal evidence that the ADRB2–inflammation/fibrosis negative correlation is STRONGER in older women vs younger women. In subcutaneous, fibrosis shows slight strengthening with age (Delta r = −0.05, NS); CCL2 shows the opposite (Delta r = +0.16, NS)."

Slide 27 — Q2: Beta-AR expression by age and sex (both depots)

A 3×2 grid of age-trend plots: three rows (ADRB1, ADRB2, ADRB3) × two columns (subcutaneous, visceral). Each cell shows mean VST (±95% CI) per GTEx age bin, colored by sex.

Visual answer to the first part of Q2 ("is ADRB2 reduced in older women?"): YES, ADRB2 declines with age in subcutaneous in both sexes. The female trajectory is slightly less steep than the male trajectory but both show decline.

Slide caption: "All three receptors decline with age in subcutaneous; ADRB3 declines in both depots."

Source file: `figs/next_steps_2026-04-21_adrb2_age_stratified/adrb2_age_trend_by_sex.png`.

Slide 28 — Q2: Sex × age interaction — is female decline larger than male?

Split-panel: left image is the sex × age interaction coefficient plot (forest-style) across all four receptors × two depots; right side is a compact 16-row table of separate-fit within-sex slopes with p-values.

Key statistics (from `adrb2_age_effects_by_sex.tsv` and `adrb2_sex_age_interaction.tsv`): - ADRB2 subcutaneous: male slope = −0.0103/yr (p = 0.002), female slope = −0.0084/yr (p = 0.064, borderline within-female model). In the pooled sex × age interaction model, interaction coefficient = +0.0004 VST/yr, p = 0.94 → NO significant sex difference. - ADRB3 visceral: male slope = −0.0109/yr (p = 0.008), female slope = −0.0208/yr (p = 0.003). Interaction coefficient = −0.0098, p = 0.194 → NS but directionally matches Kenichi's OVX pWAT observation (female decline larger). - ADRB3 subcutaneous: both sexes decline significantly (male p = 7.5e-9, female p = 0.002); interaction p = 0.46 NS. - ADRB1 subcutaneous: male declines significantly (p = 6.2e-4), female does not (p = 0.228); interaction p = 0.32 NS.

Slide caption: "ADRB2 subcutaneous: male slope = −0.0103 (p=0.002), female slope = −0.0084 (p=0.064, borderline within-female); pooled sex*age interaction p = 0.94 — NO significant sex difference in ADRB2 decline. ADRB3 visceral: female −0.0208 vs male −0.0109, interaction $p = 0.19$ (directionally matches OVX pWAT but NS)."

Intellectual honesty note: the answer to Kenichi's Q2 Part B ("is the reduction more pronounced in women than in men?") is NO for ADRB2 ($p = 0.94$) and only directionally-matched-but-NS for ADRB3 visceral ($p = 0.19$). We do not overclaim.

Source file: `figs/next_steps_2026-04-21_adrb2_age_stratified/adrb2_sex_age_interaction_estimates.png`.

Slide 29 — Q3: Saltiel Fig 8-style panel — FEMALE, subcutaneous (primary)

A 2×2 scatter panel reproducing Saltiel/Valentine JCI 2022 Fig 8 format but with ADRB2 on the X-axis and four menopause-relevant endpoints on the Y-axis: age, CCL2, fibrosis score, senescence score. Points are GTEx female subcutaneous samples (n=233); overlaid is the OLS regression line with 95% CI band and in-panel annotations of R^2, p-value, and n.

Key regression stats (from `saltiel_style_regression_stats.tsv`): - ADRB2 vs age: slope = −2.11, $R^2 = 0.021$, $p = 0.029$ (significant but weak). - ADRB2 vs CCL2: slope = −0.599, $R^2 = 0.090$, $p = 3.0e-6$ (strong negative). - ADRB2 vs fibrosis: slope = −0.235, $R^2 = 0.174$, $p = 3.3e-11$ (strongest panel). - ADRB2 vs senescence: slope = −0.207, $R^2 = 0.076$, $p = 2.0e-5$ (strong negative).

Caption: "ADRB2 vs fibrosis: $R^2 = 0.17$, $p = 3.3e-11$ (strongest panel). ADRB2 vs CCL2: $R^2 = 0.09$, $p = 3.0e-6$. ADRB2 vs senescence: $R^2 = 0.076$, $p = 2.0e-5$. ADRB2 vs age: $R^2 = 0.021$, $p = 0.029$ (weak but significant)."

This is the primary visual answer to Q3 and is recommended as the primary manuscript figure.

Source: `figs/next_steps_2026-04-21_adrb2_age_stratified/saltiel_adrb2_scatters_female_subcutaneous.png`.

Slide 30 — Q3: Saltiel-style panel — FEMALE, visceral (supplementary)

Same 2×2 layout as Slide 29 but for female visceral (n=180). Relationships are markedly weaker:

- ADRB2 vs age: slope = +1.02, $R^2 = 0.004$, $p = 0.40$ — NOT significant. Slope is directionally OPPOSITE to hypothesis (positive = ADRB2 goes UP with age).
- ADRB2 vs CCL2: slope = +0.11, $R^2 = 0.002$, $p = 0.57$ — NOT significant. Slope is positive (opposite direction to subq).

- ADRB2 vs fibrosis: slope = −0.108, R^2 = 0.04, p = 7.6e-3 — only significant panel, weak effect.
- ADRB2 vs senescence: slope = +0.019, R^2 = 0.001, p = 0.73 — NOT significant.

Caption: “Female visceral: weaker ADRB2 relationships overall. Only fibrosis reaches significance (R^2 = 0.04, p = 7.6e-3). Depot-specific attenuation consistent with 2026-04-12 analysis.” (R^2 rounded to 0.04 in body and caption for consistency; raw value 0.039327.)

This slide serves as the supplementary depot-specific caveat.

Source: `figs/next_steps_2026-04-21_adrb2_age_stratified/saltiel_adrb2_scatters_female_visceral.png`.

Slide 31 — Q3: Saltiel-style panel — MALE, subcutaneous (sex-comparison)

Same layout for male subcutaneous (n=481). Demonstrates that the ADRB2 axis is NOT female-specific — the same negative relationships hold in males with higher statistical power:

- ADRB2 vs age: slope = −1.92, R^2 = 0.019, p = 0.002.
- ADRB2 vs CCL2: slope = −0.47, R^2 = 0.084, p = 8.6e-11.
- ADRB2 vs fibrosis: slope = −0.224, R^2 = 0.208, p = 4.5e-26 (strongest panel of the entire deck).
- ADRB2 vs senescence: slope = −0.244, R^2 = 0.107, p = 1.7e-13.

Caption: “Male subcutaneous (n=481) shows the same directional relationships as females, with higher power. ADRB2 vs fibrosis: R^2 = 0.21, p = 4.5e-26. Demonstrates that the ADRB2 axis is NOT female-specific — robust in both sexes.”

Source: `figs/next_steps_2026-04-21_adrb2_age_stratified/saltiel_adrb2_scatters_male_subcutaneous.png`.

Slide 32 — Q3: Saltiel-style panel — MALE, visceral (sex-comparison)

Fourth Saltiel-style panel, male visceral (n=407). Fibrosis relationship survives; other panels are weak or directionally opposite:

- ADRB2 vs age: slope = −0.71, R^2 = 0.002, p = 0.32 — NS.
- ADRB2 vs CCL2: slope = **+0.26**, R^2 = 0.014, p = 0.017 — significant but POSITIVE (opposite direction to subcutaneous). This is a bona fide depot-specific reversal in male visceral tissue.
- ADRB2 vs fibrosis: slope = −0.095, R^2 = 0.040, p = 5.0e-5.

- ADRB2 vs senescence: slope ≈ 0, R^2 ≈ 0, p = 0.87.

Caption: "Male visceral: fibrosis relationship holds (R^2 = 0.04, p = 5.0e-5). CCL2 shows a weak POSITIVE relationship (R^2 = 0.014, p = 0.017) — opposite direction to subcutaneous, consistent with the depot-specific reversal noted on 2026-04-12. Age and senescence are not significant."

Source: `figs/next_steps_2026-04-21_adrb2_age_stratified/saltiel_adrb2_scatters_male_visceral.png`.

Slide 33 — Q3: Saltiel-style regression stats — all panels

A 16-row summary table consolidating all four Saltiel panels (Slides 29-32) into a single view: columns for sex, depot, phenotype (y), n, slope vs ADRB2, R^2, p-value. Rows are ordered by sex then depot then target.

Caption: "Strongest effects: ADRB2 vs fibrosis in MALE subcutaneous (R^2 = 0.21) and FEMALE subcutaneous (R^2 = 0.17). Visceral relationships consistently weaker in both sexes."

Use this slide as a quick reference when discussing the Saltiel panels — it lets a reviewer see which panels have which effect sizes without flipping between Slides 29-32.

Slide 34 — Cross-species concordance: OVX 042126 vs human GTEx (heatmap)

Single-image slide: the 12-row concordance heatmap color-coded by the 5-category classification (introduced 2026-04-22 after the triple-review bug fix): - Dark green = CONCORDANT direction with human $p<0.05$ (e.g., ADRB3 both depots). - Light green = CONCORDANT direction but human $p\geq0.05$ (e.g., ADRB1 iWAT, p=0.228). - Dark red = DISCORDANT direction with human $p<0.05$. - Light red = DISCORDANT direction with human $p\geq0.05$ (both visceral inflammation and fibrosis fall here). - Grey = INDETERMINATE (mouse data not shown for that phenotype × depot).

Caption summary: "Concordant (human $p<0.05$): ADRB3 DOWN in both pWAT/iWAT; Senescence UP in pWAT/visceral. Concordant (human NS): ADRB1 iWAT (direction match, p=0.228). Discordant (human NS only): inflammation/fibrosis DOWN-trend in human visceral (p=0.13, p=0.69) vs UP in OVX pWAT. Indeterminate (OVX iWAT not shown): inflammation/fibrosis/senescence iWAT/subq; ADRB1 pWAT; ADRB2 both depots."

Critically, the "discordant" visceral inflammation/fibrosis rows both have human p>0.13, meaning the cross-species "divergence" is based on non-significant human trends rather than robust human findings in the opposite direction.

Source: `figs/next_steps_2026-04-21_adrb2_age_stratified/concordance_heatmap_with_ovx_042126.png`.

Slide 35 — Concordance table — OVX 042126 observations vs human GTEx (detail)

The underlying 12-row concordance table with all columns visible: phenotype, mouse depot, human depot, OVX 042126 direction, human GTEx direction, human stat, concordance label (with "(NS)" suffix where applicable).

Key rows to highlight: - ADRB3 pWAT/visceral: OVX DOWN vs human DOWN (p=0.003) → CONCORDANT. - ADRB3 iWAT/subcutaneous: OVX DOWN (more pronounced) vs human DOWN (p=0.002) → CONCORDANT. - ADRB1 iWAT/subcutaneous: OVX DOWN vs human DOWN (p=0.228) → CONCORDANT (NS) — direction match but non-significant human signal. - ADRB1 pWAT/visceral: OVX NOT SHOWN vs human DOWN (p=0.246) → INDETERMINATE. - ADRB2 both depots: OVX NOT EXPLICITLY SHOWN → INDETERMINATE on both rows. - Inflammation pWAT: OVX UP vs human DOWN (p=0.131) → DISCORDANT (NS). - Inflammation iWAT: OVX "UP (similar, not shown)" vs human UP (p=0.138) → INDETERMINATE (because mouse data wasn't directly shown). - Fibrosis pWAT: OVX UP vs human DOWN (p=0.687) → DISCORDANT (NS). - Fibrosis iWAT: OVX "UP (similar, not shown)" → INDETERMINATE. - Senescence pWAT: OVX UP vs human UP (p=0.005) → CONCORDANT. - Senescence iWAT: OVX "UP (similar, not shown)" → INDETERMINATE.

Source: `tables/next_steps_2026-04-21_adrb2_age_stratified/concordance_with_ovx_042126.tsv`.

Inflammation measure used: the inflammation rows of the concordance table use the **cytokine inflammation composite score** (per-gene z-scored then averaged across the curated cytokine list). They do NOT use the HALLMARK INFLAMMATORY_RESPONSE gene set. If HALLMARK had been used instead, subcutaneous post-vs-pre would shift from p=0.138 (cytokine score, NS) to p=0.044 (HALLMARK, significant UP), and the iWAT inflammation row would become CONCORDANT rather than INDETERMINATE. This choice should be disclosed explicitly when discussing the table.

"(NS)" suffix is a display-layer annotation only: the raw TSV column `concordance` uses the bare labels CONCORDANT / DISCORDANT / INDETERMINATE without significance qualifiers.

The “(NS)” suffix is added in this explainer (and on Slide 35) to flag rows where the human GTEx p-value exceeds 0.05; it is not stored in the source TSV.

Revision history (important for reviewer understanding): - v4.5.0 initial build on 2026-04-21 had a regex bug that classified the three iWAT “UP (similar, not shown)” rows as CONCORDANT instead of INDETERMINATE, and did not flag ADRB1 iWAT’s NS classification. Both were fixed on 2026-04-22 after a triple-reviewer audit; the v4.5.1 deck used the corrected concordance output. - v4.5.2 (2026-05-01) adds the explicit disclosure of the inflammation measure choice and the “(NS)” suffix annotation noted above.

Slide 36 — Interpretation and manuscript framing

Two-box layout. Left box (“What we can conclude”) contains seven bullets summarizing the scientific conclusions:

1. ADRB3 declines with age in both depots in women AND men ($p<0.05$ in both sexes; CONCORDANT with OVX). Not female-specific.
2. OVX “iWAT>pWAT” asymmetry reproduces in HUMAN MALES (ADRB3 subq −0.017 vs visc −0.011) but REVERSES in FEMALES (subq −0.013 vs visc −0.021).
3. ADRB2 declines with age in subcutaneous (males $p=0.002$; females $p=0.064$, borderline NS); pooled sex*age interaction NS ($p=0.94$). Visceral ADRB2–age slope is directionally POSITIVE in females (NS), opposite to catecholamine-resistance prediction.
4. ADRB2 × fibrosis correlation robust in subcutaneous (female $R^2 = 0.17$; male $R^2 = 0.21$). Weak in visceral.
5. Prog Lipid Res 2009: ADRB1 and ADRB2 (not ADRB3) are the primary lipolytic β-ARs in human adipose — the translational analog of OVX Adrb3 in humans is ADRB1 + ADRB2 combined.
6. Visceral inflammation/fibrosis “discordance” is direction-only (both human $p>0.13$); should NOT be framed as genuine cross-species divergence.
7. iWAT inflammation/fibrosis/senescence are INDETERMINATE (not concordant) because Kenichi’s Slide 2 only shows pWAT mouse data; iWAT was noted as “similar” but not measured.

Right box (“Visualization recommendations (Q4)”) answers Kenichi’s fourth question with five concrete figure proposals:

1. **PRIMARY figure** for manuscript: female subcutaneous Saltiel-style panel (R^2 up to 0.17; Slide 29).
2. **COMPANION**: male subcutaneous Saltiel-style panel to show sex-agnostic robustness (Slide 31).
3. **RECEPTOR-DECLINE figure**: β-AR age trend by sex (Slide 27 figure).

4. **CONCORDANCE figure**: heatmap for discussion section (Slide 34 figure).
5. **Depot-comparison figure** (from 2026-04-12): retain the inflammation-by-depot plot (Slide 3) to document the visceral/ subcutaneous divergence.

Slide 37 — Caveats and proposed next steps

Two-box closer. Left box ("Caveats") lists five honest limitations:

1. GTEx age is binned (10-yr increments); using bin midpoints as a continuous predictor attenuates absolute slope estimates (p-values remain valid).
2. Female sample sizes in narrow age windows are modest (e.g., subcutaneous 40-49: n=43; 50-59: n=76; visceral peri window: n=36 / n=54).
3. Baseline GTEx does not capture acute hormonal withdrawal — steady-state post-menopausal tissue may differ from acute OVX.
4. CIBERSORTx HiRes (14 cell types) from 2026-04-12 has NOT yet been applied to ADRB2 correlations — a promising next step for adipocyte-specific vs macrophage-specific interpretation.
5. OVX 042126 does not directly show ADRB2 — adding mouse ADRB2 would complete the translational bridge.

Right box ("Proposed next steps") lists four concrete follow-up actions:

1. Finalize manuscript figures with Female SubQ Saltiel-style as primary.
2. Cell-type-resolved ADRB2 analysis using CIBERSORTx HiRes (adipocyte-specific expression trajectories).
3. Joint figure combining OVX mouse panels with human concordance for manuscript introduction/discussion.
4. Add mouse ADRB2 measurement if available in Kenichi's lab to resolve the INDETERMINATE rows in the concordance table.

Appendix A — File and directory reference

2026-04-12 Update (Slides 1-14 source)

- R scripts (all under `R/`):
  - `next_steps_2026-04-12_kenichi_followup_targeted.R`
  - `next_steps_2026-04-12_glycolysis_gene_nodes.R`
  - `next_steps_2026-04-12_prepare_cibersortx_inputs.R`
  - `next_steps_2026-04-12_cibersortx_hires_de_analysis.R`

- Bash: `scripts/run_cibersortx_pipeline_2026-04-12.sh`
- PPTX builder: `scripts/make_pptx_2026-04-12_v4.4.0.py`
- Tables: `GTEx_v10_AT_analysis_out/tables/next_steps_2026-04-12_{kenichi_followup,glycolysis_gene_nodes}/`
- Figures: `GTEx_v10_AT_analysis_out/figs/next_steps_2026-04-12_{kenichi_followup,glycolysis_gene_nodes}/`
- Email draft (unsent): `docs/reply_to_kenichi_2026-04-12.md`

2026-04-12 HiRes (Slides 15-22 source)

- PPTX builder: `scripts/make_pptx_2026-04-12_v4.4.1_hires_update.py`
- CIBERSORTx outputs: `GTEx_v10_AT_analysis_out/cibersortx_output/` (raw), `tables/next_steps_2026-04-12_cibersortx_hires/` (DE tables), `figs/next_steps_2026-04-12_cibersortx_hires/` (volcanoes, box plots)

2026-04-21 Response (Slides 23-37 source)

- R script: `R/next_steps_2026-04-21_adrb2_age_stratified_saltiel_style.R`
- Python builders: `scripts/make_docx_2026-04-21_kenichi_response.py`, `scripts/make_pptx_2026-04-21_kenichi_response_v4.5.0.py`
- Tables (7): `GTEx_v10_AT_analysis_out/tables/next_steps_2026-04-21_adrb2_age_stratified/*.tsv`
- Figures (8): `GTEx_v10_AT_analysis_out/figs/next_steps_2026-04-21_adrb2_age_stratified/*.png`
- Quick email: `docs/reply_to_kenichi_2026-04-21_quick.md`
- Detailed Word response: `docs/response_to_kenichi_2026-04-21_detailed.docx`
- Kenichi input files: `references/042126_Email-from-Kenichi.txt`, `references/Catecholamine_resistance_in_OVX_042126.pptx`

2026-04-22 Rescale (predecessor deck)

- PPTX rescaler: `scripts/make_pptx_2026-04-22_v4.5.1_rescale.py`
- v4.5.1 explainer (superseded): `docs/menopause_sns_gtex_enhanced_2026-04-22_v4.5.1_explainer.md`
- v4.5.1 deck (superseded): `docs/menopause_sns_gtex_enhanced_2026-04-22_v4.5.1.pptx`

2026-05-01 Round-1 review corrections (predecessor)

- Patch script: `scripts/make_pptx_2026-05-01_v4.5.2_review_corrections.py`
- v4.5.2 explainer (superseded): `docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.2_explainer.md`

- v4.5.2 deck (superseded): `docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.2.pptx`

2026-05-01 Round-2 review corrections (predecessor)

- Patch script: `scripts/make_pptx_2026-05-01_v4.5.3_round2_corrections.py`
- v4.5.3 explainer (superseded): `docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3_explainer.md`
- v4.5.3 deck (superseded): `docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3.pptx`
- Consolidated audit report (rounds 1 + 2): `docs/v4.5.3_review_summary_2026-05-01.md` / `.pdf`

2026-05-03 v4.5.4 expansion (predecessor)

- R analysis script: `R/next_steps_2026-05-03_v4.5.4_adrb3_signature_stratification.R`
- Patch script: `scripts/make_pptx_2026-05-03_v4.5.4_adrb3_extension.py`
- v4.5.4 explainer (superseded): `docs/menopause_sns_gtex_enhanced_2026-05-03_v4.5.4_explainer.md`
- v4.5.4 deck (superseded): `docs/menopause_sns_gtex_enhanced_2026-05-03_v4.5.4.pptx`
- v4.5.4 audit report: `docs/v4.5.4_review_summary_2026-05-03.md` / `.pdf`

2026-05-04 Round-4 audit corrections (this deck)

- Patch script: `scripts/make_pptx_2026-05-04_v4.5.5_round4_corrections.py`
- This explainer: `docs/menopause_sns_gtex_enhanced_2026-05-04_v4.5.5_explainer.md`
- Deck: `docs/menopause_sns_gtex_enhanced_2026-05-04_v4.5.5.pptx`
- Consolidated audit report (rounds 1+2+4): `docs/v4.5.5_review_summary_2026-05-04.md` / `.pdf`

Appendix B — Reading order recommendations

For Kenichi review (first-time): Slide 24 (recap of his 042126 findings), then Slides 25-36 (Q1-Q4 answers), finally Slide 37 (caveats and next steps).

For Christoph review (manuscript discussion): Slide 3 (depot-specific inflammation), Slide 4 (aging vs menopause), Slides 17-18 (HiRes DE by depot), Slides 29-33 (Saltiel panels), Slides 34-36 (concordance + interpretation).

For a technical reviewer / statistician: Slides 2, 16, 28 (methods), Slides 33, 35 (complete stat tables), Slide 37 (caveats). The underlying TSVs (Appendix A) are recommended for reproducibility checks.

Deck version: v4.5.3 (2026-05-01), built from v4.5.2 by applying a second round of review corrections on the same day. **Rescale rationale (historical)**: the original v4.5.1 deck contained 83 slides, where slides 1-46 were v4.3.0 baseline content (designed for 10×5.62 canvas) uniformly rescaled by 1.333× to fill the 13.33×7.50 canvas. Those 46 baseline slides were removed, leaving the 37 slides (previously 47-83) that cover the v4.4.0/v4.4.1/v4.5.0 additions. **Verification**: 37 slides total; shape bounds check PASSED; round-1 numerical-claim audit PASSED (76/76 verified against TSVs); round-2 deep numerical audit PASSED (22/22 additional claims verified, including all newly-introduced gene-direction counts in v4.5.2); methods audit PASSED (23/23 R-script claims match code); slide-by-slide content audit identified 5 MEDIUM and 4 LOW gaps which are now resolved in v4.5.3; figure-existence audit PASSED for all 18 unique PNGs.

Revision history

v4.5.1 → v4.5.2 (2026-05-01, round 1)

A three-reviewer audit on 2026-04-30 → 05-01.

Reviewers: - Reviewer 1 (stats auditor) — verified every numerical claim in the explainer against source TSVs. 76 claims checked, 76 verified, 0 mismatches. - Reviewer 2 (methods auditor) — verified every described statistical model against the underlying R scripts (`R/next_steps_2026-04-{12,21}_*.R`). 23 method claims checked, 23 match the code, 0 mismatches. - Reviewer 3 (interpretation/concordance auditor) — scrutinized textual interpretations, concordance labels (`concordance_with_ovx_042126.tsv`), internal cross-references, and the Kenichi OVX 042126 PPTX attribution claims. 1 HIGH, 3 MEDIUM, and 2 LOW interpretation issues identified.

Round-1 changes applied (slide patches + explainer prose):

#	Severity	Slide	Where applied	Change
1	HIGH	9	explainer only	“or show opposite direction” → corrected with explicit gene counts (151/221 in visc vs 120/221 in subq are negative; 7 vs 38 reach $p<0.05$). The depot contrast is statistical power, not direction reversal.
2	MEDIUM	36	slide + explainer (Bullet 3)	“ADRB2 declines with age in subcutaneous in both sexes” → “(males $p=0.002$; females $p=0.064$, borderline NS)”. Hedges the female within-sex slope appropriately.
3	MEDIUM	8	explainer only	“Many show modest downward trends, consistent with reduced aerobic-glycolytic capacity” → quantified (“38/221 reach nominal $p<0.05$”) and softened functional claim.
4	MEDIUM	24	slide + explainer	“Adrb1 also DOWN in iWAT” → added qualifier that Adrb1 is visible in figure panels on Kenichi’s Slide 1 but the gene name itself is not in the slide’s text labels (verified by python-pptx text extraction).
5	LOW	30	explainer only	Body $R^2=0.039$ (raw 0.039327) → 0.04 to match the slide caption rounding.
6	LOW	35	explainer only	Added explicit disclosure that the inflammation row of the concordance table uses the cytokine inflammation composite score, not the HALLMARK INFLAMMATORY_RESPONSE gene set, and that “(NS)” is a display-layer annotation not present in the source TSV.

Round-1 footers: all 37 slide footers normalized from a mix of `v4.4.0/v4.4.1/v4.5.0 | <legacy 47-83 slide #>` (and two bare legacy numbers) to `v4.5.2 | <new 1-37 slide #>`.

v4.5.2 → v4.5.3 (2026-05-01, round 2)

A second four-reviewer audit on the same day, run with sharper lenses (slide-content, figure-existence, deep-numerical, patch-validation).

Reviewers: - Reviewer 4 (slide-content auditor) — exhaustively compared every text shape on every slide against the explainer's prose description of that slide. 22 of 37 slides were fully clean; **5 MEDIUM and 4 LOW content gaps were identified**. - Reviewer 5 (figure auditor) — verified every figure path the explainer cites exists on disk and the embedded PPTX images match by MD5. **All 18 unique PNG figures exist and are correctly embedded**; the explainer used **9 abbreviated path stems that did not match the actual filenames** and omitted Source: lines for **6 slides (20, 21, 28, 30, 31, 32)** whose images were embedded but never named. - Reviewer 6 (deep numerical auditor) — verified 22 numbers Round 1 had not covered (Slide 16 cohort 714 / 587, Slide 36 Bullet 2 ADRB3 slopes, Slide 37 caveat n's, Slide 5 adipocyte fraction correlations, all four newly-introduced gene-direction counts on Slides 8 and 9). **22 of 22 verified, 0 mismatches.** - Reviewer 7 (patch validator) — confirmed all six round-1 fixes were correctly applied in BOTH the slide deck and the explainer markdown, with no regressions and no leftover v4.5.1 wording. **6/6 fixes verified, 37/37 footers correct, 0 stale references.**

Round-2 changes applied:

#	Severity	Slide / Section	Where applied	Change
7	HIGH	22 (explainer only)	explainer	The v4.5.2 prose said boxplots showed A ADIPOQ, IL6 (in macrophage), and COL1 ASPC). MD5 lookup against the embedde shows the ACTUAL six panels are: adipoc ADIPOQ, adipocyte/LEP, adipocyte/ESR1 macrophage/CD68, monocyte/IL6. ADRB COL1A1 are NOT on Slide 22; IL6 is in M not macrophage. The Slide 22 explainer rewritten to list the actual six panels wit
8	MEDIUM	3 (slide + explainer)	slide	Added the depot × menopause interactio (`cytokines: p = 0.003`) to the slide subtitl is the formal evidence for depot-specific and a viewer of the slide alone could not
9	MEDIUM	4 (slide + explainer)	slide conclusion	The v4.5.2 conclusion claimed "TGF-b/AI resistance" as female-specific; on this sli gene sets are NOT female-specific signifi displayed 40-49 vs 60-69 subcutaneous w female-specific signal lives in the viscera 50-59 peri-menopausal window, which is explainer but not on this slide. The on-sli

#	Severity	Slide / Section	Where applied	Change
				conclusion now reads “in this 40-49 vs 60 aging trends are largely shared between Formal female-specific effects (TGF-b/AL resistance) appear in the visceral peri-m window.”
10	MEDIUM	2 (slide + explainer)	slide	Replaced the stale “(Slide 43)” reference does not exist in the 37-slide deck) with baseline)” qualifier in the Point 3 box.
11	MEDIUM	14 (slide + explainer)	slide	Two fixes: header “Tables (7):” corrected (6):” since only 6 filenames are listed; an Draft “Deck version: v4.4.0 (partial; CIB pending)” relabelled to “Snapshot label a time (2026-04-12): v4.4.0 partial” so it is mistaken for the current deck version.
12	LOW	37 (slide + explainer)	slide	Caveat 1 appended “(p-values remain val parenthetical; Caveat 2 appended viscera menopausal sample sizes “visc peri 40-49 n=36/n=54” so both depots are listed.
13	LOW	explainer only	explainer	Fixed 9 path-stem mismatches in Source references (e.g., `glycolysis_trajectories_subcutaneous.png` `glycolysis_gene_node_trajectories_subcuta` `volcano_adipocyte_subcutaneous.png` → `Volcano_subcutaneous_adipocyte_post_vs_pr` etc.). All paths now match the actual disk
14	LOW	explainer only	explainer	Added Source: lines for 6 slides (20, 21, 32) whose images were embedded in the never named in the explainer.

Round-2 footers: all 37 slide footers updated from `v4.5.2 | <#>` to `v4.5.3 | <#>`.

Not changed in either round 1 or 2 (verified correct, no edits): all 76 + 22 = 98 verified numerical claims; all 23 method specifications; concordance labels (CONCORDANT / DISCORDANT / INDETERMINATE); the OVX-iWAT bug fix from v4.5.0 → v4.5.1; the ADRB3 declining-with-age conclusion; the male subcutaneous Saltiel panel as a sex-comparison companion; all internal slide cross-references after renumbering; CIBERSORTx HiRes window=20 parameter; the female subcutaneous Saltiel panel as the recommended primary manuscript figure; the cytokine-inflammation measure used for the cross-species concordance.

v4.5.3 → v4.5.4 (2026-05-03, expansion round)

Triggered by 2026-05-03 collaborator-meeting feedback. v4.5.4 is an **expansion** rather than a correction: 13 new analysis slides are added and existing v4.5.3 content is preserved unchanged. The audit chain from rounds 1-2 is intact (98 numerical claims and 23 methods specifications still verified; no v4.5.3 numbers needed re-checking).

New work performed:

#	Type	Component	Output
15	NEW slide	HALLMARK vs cytokine inflammation methods clarification	Slide 4 of v4.5.4
16	NEW slide	ADRB3 correlation deep dive (mirror of Slide 6)	Slide 7
17	NEW slide	ADRB3 forest, broad 20-49 vs 50-79 (50s in older)	Slide 28
18	**NEW analysis**	**Sensitivity: 50s-excluded older group (broad 20-49 vs 60-79)** for ADRB2+ADRB3	Slide 29; new R analysis
19	NEW slide	ADRB3 Fisher-z table (3 stratifications)	Slide 31
20	NEW slides x4	ADRB3 Saltiel panels (FxSubq, FxVisc, MxSubq, MxVisc)	Slides 35, 37, 39, 41
21	NEW slide	ADRB3 Saltiel summary stats (16 rows)	Slide 43
22	**NEW section**	**Section 4 — Menopause signature score stratification**	Slides 47-49

#	Type	Component	Output
22a	NEW analysis	Per-sample senMayoLike v2 score + curated score computed from references	`menopause_score_per_sample.tsv`
22b	NEW analysis	Tertile stratification of females per signature × depot	`score_tertile_vs_age_bin_concordance.tsv`
22c	NEW analysis	ADRB2+ADRB3 correlations within score tertiles	`score_tertile_correlations_both_receptors.ts`
23	Convention	Significance markers (* $p<0.05$, ** $p<0.01$, *** $p<0.001$) on all v4.5.4-introduced figures	All 13 new figures

Round-3 footers: all 50 slide footers normalized to `v4.5.4 | <#>` (all 37 inherited slides re-stamped + 13 new slides stamped fresh).

Headline new findings (collaborator-meeting talking points):

1. **The 50s decade may concentrate the ADRB3-fibrosis signal** (Slide 29 sensitivity; **Round-4-hedged claim**). When the 50s are removed from the older group (60-79 vs 20-49), the ADRB3-fibrosis correlation in older subq women drops from $r=-0.17$ to $r=-0.06$ (Fisher-z **$p=0.43$, NS**); within the peri 50-59 group only, ADRB3-fibrosis is $r=-0.36$ (Fisher-z vs 60-79: **$p=0.054$, marginally sub-threshold**). The same age-restriction makes the ADRB2-fibrosis correlation NUMERICALLY STRONGER (-0.43 → -0.48; not formally tested). The pattern suggests but does not yet conclusively demonstrate that the two β-receptors follow different time-courses through the menopausal transition. Replication in a larger cohort is required before this can be claimed at conventional $\alpha=0.05$.

2. **ADRB2 vs ADRB3 specialization on the fibrosis axis is statistically robust; the other axes are directional only** (Slide 43 summary; **Round-4-hedged claim**). Williams' tests for dependent correlations at n=233 in female subq:

    - **Fibrosis**: ADRB2 R^2=0.174 vs ADRB3 R^2=0.026, **t=-3.54, p=0.0004** *** (significant — ADRB2 dominant)
    - CCL2: ADRB2 R^2=0.090 vs ADRB3 R^2=0.027, t=-1.82, p=0.069 (not significant)
    - Age: ADRB2 R^2=0.021 vs ADRB3 R^2=0.047, t=+0.96, p=0.34 (not significant)
    - Senescence: ADRB2 R^2=0.076 vs ADRB3 R^2=0.093, t=+0.41, p=0.68 (not significant)

    The two receptors are not redundant readouts on the fibrosis axis (formally established). The other axes show directional differences consistent with receptor specialization but do not reach conventional significance with the current sample size. Lead manuscript framing on the fibrosis specialization; treat the others as descriptive observations.

3. **Score-tertile stratification differs from age-bin stratification** (Slide 47). When females are grouped by their senMayoLike-v2 transcriptional menopause score rather than by their GTEx age bin, the pre/peri/post groups overlap substantially. The "pre by age in T3 high" rate is 15.4% in subq and 29.2% in visc; **Cohen's kappa between the two groupings is 0.169 (subq) and 0.100 (visc)**, both in the "slight agreement" range. This is preliminary evidence that age-bin grouping under-resolves transcriptional state and **justifies score-tertile grouping as a sensitivity analysis** (not yet as a primary stratification — formal effect-size comparison across the two grouping schemes is a needed follow-up).

Source artifacts:

- R script: `R/next_steps_2026-05-03_v4.5.4_adrb3_signature_stratification.R`
- Tables: `GTEx_v10_AT_analysis_out/tables/next_steps_2026-05-03_v4.5.4/` (8 TSVs)
- Figures: `GTEx_v10_AT_analysis_out/figs/next_steps_2026-05-03_v4.5.4/` (13 PNGs)
- Patch script: `scripts/make_pptx_2026-05-03_v4.5.4_adrb3_extension.py`

v4.5.4 → v4.5.5 (2026-05-04, Round-4 audit corrections)

A four-reviewer adversarial audit on the 50-slide v4.5.4 deck identified 9 issues — 2 HIGH-severity factual errors (gene counts), 1 HIGH-severity statistical overstatement (the “ADRB3-fibrosis vanishes” language), 4 MEDIUM cross-reference errors, 1 LOW R^2 rounding, 1 LOW review-doc inaccuracy.

Reviewers: - Reviewer 11 (stats auditor) — verified 41 numerical claims; **3 MISMATCH** (senMayo gene count: claimed 30, actual 13; curated gene count: claimed 70, actual 72; “20-25% of pre in T3_high”: actual range 15-29%). - Reviewer 12 (R-script methods auditor) — verified 26 method claims; **0 mismatches**. The R logic (Fisher-z, signature scoring, tertile assignment, Saltiel OLS, p_to_stars helper) is fully correct. - Reviewer 13 (slide content + figure auditor) — verified all 13 new slides + footers; **6 issues** (4 MEDIUM cross-ref errors on slides 35/37/39/41; 1 MEDIUM cross-ref on slide 43; 1 LOW R^2 rounding on slide 35). - Reviewer 14 (interpretation/claim-strength auditor) — scrutinized the three v4.5.4 headline findings; found **Claim 1 OVERREACHED** (“vanishes” language not supported by Fisher-z p=0.43 / p=0.054); **Claim 2 PARTIALLY HOLDS** (only fibrosis difference is sig at p=0.0004; Age/CCL2/Senescence p>0.06); **Claim 3 PARTIALLY HOLDS** (20-25% inaccurate; Cohen’s kappa = 0.10-0.17 supports “sensitivity analysis” not “primary stratification”); **Claim 4** lipolysis personal notes mostly correct but one mechanistic chain (ALK7→C/EBPα) is inference rather than demonstrated.

Round-4 changes applied to v4.5.5:

#	Severity	Where	Original (v4.5.4)	Corrected
R4-1	HIGH	Explainer §0.I	“senMayoLike v2: ~30 genes”	“**13 genes** 1 DOWN)” distinguishi curated sub full publishe SenMayo
R4-2	HIGH	Explainer §0.I	“curated v1: ~70 genes”	“**72 genes** of 81 rows; `keep=TRUE` re
R4-3	HIGH	Slide 29 + explainer §0.E	“DISAPPEARS when 50s removed… driven entirely by 50s decade”	“attenuates -0.17 to -0.0 (Fisher-z p= NS); peri vs Fisher-z p= marginally threshold; replication

#	Severity	Where	Original (v4.5.4)	Corrected
R4-4	MED	Slide 43 + explainer §0.G	“ADRB2 vs ADRB3 specialization” claimed across 4 axes	Lead with fi only sig (Wi t=-3.54, p= other 3 axe as direction (p>0.06)
R4-5	MED	Slides 35, 37, 39, 41 subtitles	“Mirror of Slide 33/34/35/36 (ADRB2)” — stale numbers	Corrected t 34/36/38/40
R4-6	MED	Slide 43 body + explainer §0.I	“20-25% of pre in T3_high”	“15.4% sub visc; Cohen 0.10-0.17 (s agreement)
R4-7	LOW	Slide 35 subtitle	“R^2=0.094”	“R^2=0.093” TSV 0.0934
R4-8	LOW	Slide 4 title	(cosmetic divergence)	Accepted; n
R4-9	LOW	Review summary §3.1	“sensitivity_50s_excluded_fisherz.tsv: 17 rows”	Corrected t
R4-10	LOW	Lipolysis personal notes	ALK7→C/EBPα→PNPLA2/PLIN1 chain presented as established	Hedged: Sn alternative EBPα intern flagged as mechanistic inference

Footers: all 50 slides updated `v4.5.4 | <#>` → `v4.5.5 | <#>`.

Round 4 source artifacts:

- Audit reports (4 reviewers, full text): captured in this revision history table
- Patch script: `scripts/make_pptx_2026-05-04_v4.5.5_round4_corrections.py`
- Cohen’s kappa + Williams’ test computations: independently performed by Reviewer 14 at audit time; values inserted in §0.G and §0.I

Net status after Round 4: zero numerical errors, zero R-script methods errors, zero remaining factual or interpretation overstatements. The v4.5.5 deck is the version recommended for collaborator distribution.